

Pre-Algebra Test II

Instructions

- i) As always, write a title, name, date, and course.
- ii) Number each problem, circle the number, and show all work.
- iii) Draw a rectangle around your answers.
- iv) Rewrite the equation whenever you perform an operation on both sides to preserve the result of the previous step.

Solve.

① $x + 6 = -3$ ② $-5 + y = -11$

③ $1 = \frac{z}{-7}$ ④ $8x = 4$

⑤ $7 = \frac{y}{5} - 1$ ⑥ $2z + 10 = -3$

⑦ $-7x + 3 = 1$ ⑧ $-10 = \frac{y}{2} - 5$

⑨ $7z - 9z + 14 = 15$ ⑩ $-6 = -9x + 8x - 7$

⑪ $-2y - 4y + 6 = 6$ ⑫ $-2 = -10x + 20x + 5$

$$\textcircled{13} \quad -7y + 6y - 2 + 1 = 4 - 9 + 3$$

$$\textcircled{14} \quad -1 - 9 - 4 + 10 = 8 + 3z + (-1) + 2z - 5 - 3z$$

$$\textcircled{15} \quad -4 = 2x - x - 4 \quad \textcircled{16} \quad \frac{5y}{6} - 2y + 1 = -3 + 11$$

$$\textcircled{17} \quad -4 - 6 + (-3) = 10z + 10z + 7$$

$$\textcircled{18} \quad -x + 5x - 8 + 2 = 8 - 12 - 3$$

$$\textcircled{19} \quad -1 - 1 - 2 = 4 - \frac{y}{3} \quad \textcircled{20} \quad 2 - 9z + 5z - 3 + z = -3 - 4$$

$$\textcircled{21} \quad -5 + 6 + 4x + 2 + (-8x) = 7 - 8 - 10 + 5 + (-7)$$

$$\textcircled{22} \quad \frac{3y}{8} - \frac{7y}{8} + \frac{y}{2} - 9 + (-2) + \frac{3y}{2} = 10 - 1 - 3$$

$$\textcircled{23} \quad -3 + 5 - 1 - 6 + 4 = -\frac{x}{4} + \frac{x}{2} + 1 - \frac{3x}{4} + 3 - 1$$

$$\textcircled{24} \quad -\frac{z}{3} + \frac{5z}{6} + 6z - 8 = 2 - 9 - 1 + 3$$

$$\textcircled{25} \quad -3 + (-3) + 12 = -\frac{3x}{5} - \frac{3x}{10} - \frac{x}{5} + (-8) - 7 + 11 + \frac{7x}{10}$$

Pre-Algebra Test II Answers

① $x = -9$

② $y = -6$

③ $z = -7$

④ $x = \frac{1}{2}$

⑤ $y = 40$

⑥ $z = -13\frac{1}{2}$

⑦ $x = 2\frac{1}{7}$

⑧ $y = -10$

⑨ $z = -\frac{1}{2}$

⑩ $x = -1$

⑪ $y = 0$

⑫ $x = -7\frac{1}{10}$

⑬ $y = 1$

⑭ $z = -3$

⑮ $x = 0$

⑯ $y = -6$

⑰ $z = -1$

⑱ $x = -\frac{1}{4}$

⑲ $y = 24$

⑳ $z = 2$

㉑ $x = 4$

㉒ $y = 34\frac{1}{3}$

㉓ $x = 8$

㉔ $z = 6\frac{1}{13}$

㉕ $x = -50$